🔑 **Answer Key — Teacher Copy · Lesson 5-1**

MSTAR PRACTICE WORKSHEET

Lesson 5-1: Determine the Area of Parallelograms and Rhombuses

Grade 6 Mathematics · Standard 6.GR.1

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: C

What is the area of a parallelogram with base 10 cm and height 7 cm?

A. 17 sq cm

B. 34 sq cm

C. 70 sq cm

D. 70 cm

A = b × h = 10 × 7 = 70 square centimeters.

Question 2 SELECTED RESPONSE — CORRECT: A

A parallelogram has an area of 54 sq ft and a base of 9 ft. What is the height?

A. 6 ft

B. 27 ft

C. 45 ft

D. 63 ft

A = b × h → 54 = 9 × h → h = 54 ÷ 9 = 6 ft.

Question 3 SELECTED RESPONSE — CORRECT: C

What is the area of a parallelogram with base 9 cm and height 5 cm?

A 14 sq cm

B 22.5 sq cm

C 45 sq cm

D 90 sq cm

A = b × h = 9 × 5 = 45 square centimeters.

Question 4 SELECTED RESPONSE — CORRECT: C

Which measurement is the height of a parallelogram?

A The length of the slanted side

B The perimeter divided by 4

C The perpendicular distance between the base and the opposite side

D The longest side

The height is always the perpendicular (straight up-and-down) distance between the base and the opposite side, not the slanted side.

Question 5 SELECTED RESPONSE — CORRECT: D

Cut the slanted end off a parallelogram and slide it to the other side. What shape forms, and why does that show A = base × height?

A A bigger parallelogram, so the area gets larger

B A triangle, so the area is cut in half

C A square, so all four sides must be equal

D A rectangle with the same base and height, so its area equals base x height

Sliding the cut triangle turns the parallelogram into a rectangle that keeps the same base and height. A rectangle's area is base x height, so the parallelogram's area is base x height too.

Question 6 SELECTED RESPONSE — CORRECT: C

A blueprint parallelogram has a base of 8 ft, a height of 5 ft, and a slanted side of 7 ft. What is its area?

A. 26 sq ft

B. 35 sq ft

C. 40 sq ft

D. 56 sq ft

Area = base x height = 8 x 5 = 40 sq ft. The 7 ft slanted side is not used; only the perpendicular height belongs in the formula.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: D

A parallelogram-shaped patio has a base of 16 feet and a perpendicular height of 9 feet. What is its area?

A. 25 square feet

B. 50 square feet

C. 72 square feet

D. 144 square feet

Area = base × height = 16 × 9 = 144 square feet.

- **A:** That adds 16 and 9. Area multiplies the two measurements.
- **B:** That is the perimeter, 2 times 16 plus 9. Area multiplies base by height.
- **C:** That halves the product, which is the TRIANGLE rule. A parallelogram is base times height with no halving.

Part B — correct answer: D

The slanted side of that patio measures 11 feet. Why is 11 feet NOT used in the area calculation?

A. Because 11 is an odd number.

B. Because the slanted side is always longer than the height.

C. Because only the base matters in a parallelogram.

D. Because the height must be perpendicular to the base, and the slanted side is not.

Area = b · h requires the PERPENDICULAR height. The slanted side is longer than the perpendicular distance between the parallel bases, so using it would overstate the area.

Question 8 SELECT ALL THAT APPLY — CORRECT: A, B, D, E

Select ALL statements that are true about finding the area of a parallelogram:

- **A** Any side can be chosen as the base, as long as the height is perpendicular to that side.
- **B** The height is the distance straight across between the two parallel bases.
- **C** The slanted side can be used as the height.
- **D** A rectangle is a parallelogram, so A = b · h works for it too.
- **E** Doubling only the base doubles the area.

A, B, D, and E are true. C is false: the slanted side is longer than the perpendicular height, so using it overstates the area.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

A parallelogram has a base of 12 cm. If you double the height, the area doubles too. Explain why this happens using the formula, and give a specific example with numbers.

Model answer: In A = b × h the base of 12 cm is a fixed factor, so the area is just 12 times whatever the height is. With h = 5, A = 12 × 5 = 60 sq cm; doubling to h = 10 gives A = 12 × 10 = 120 sq cm, exactly double, because doubling one factor doubles the product.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.